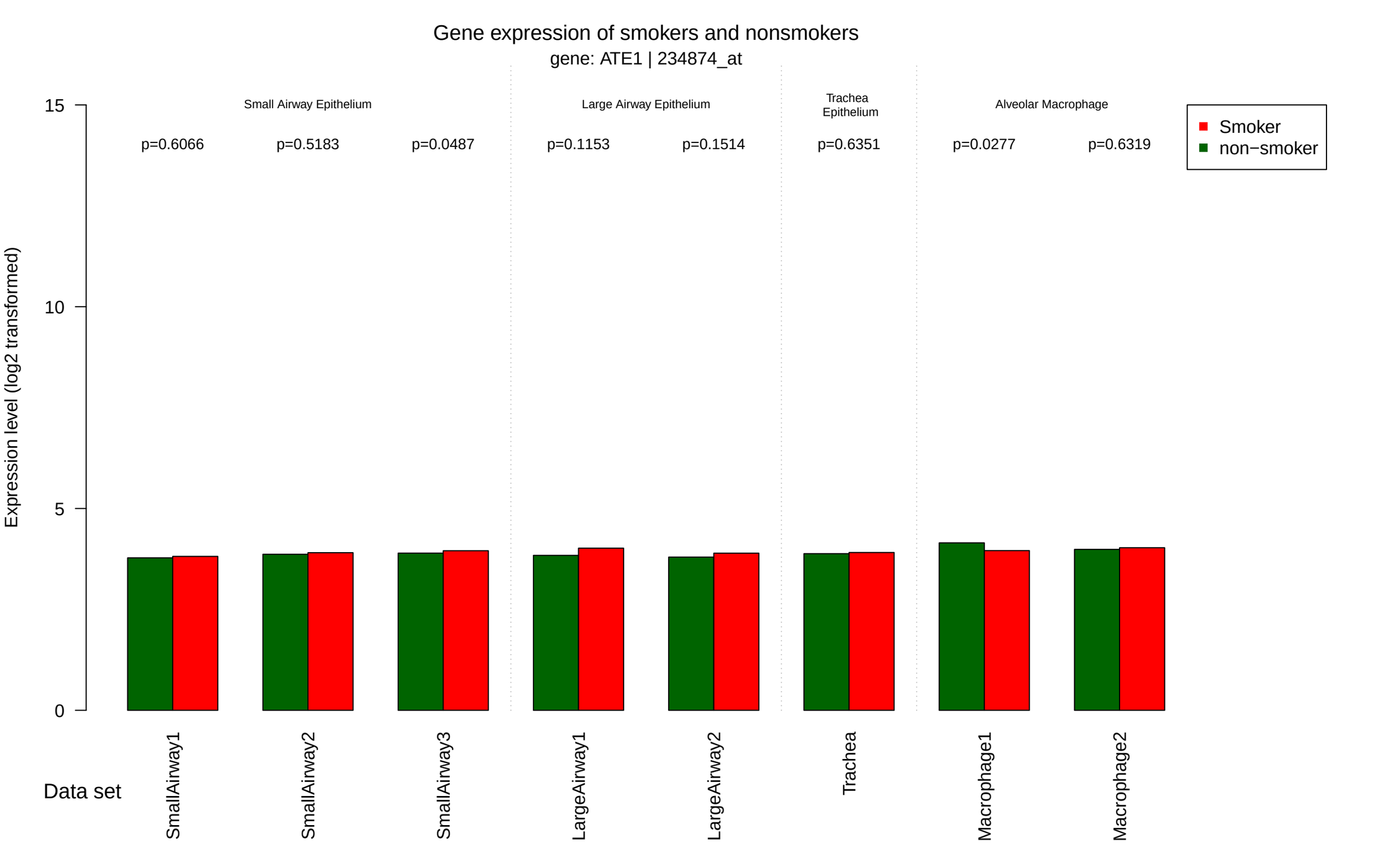
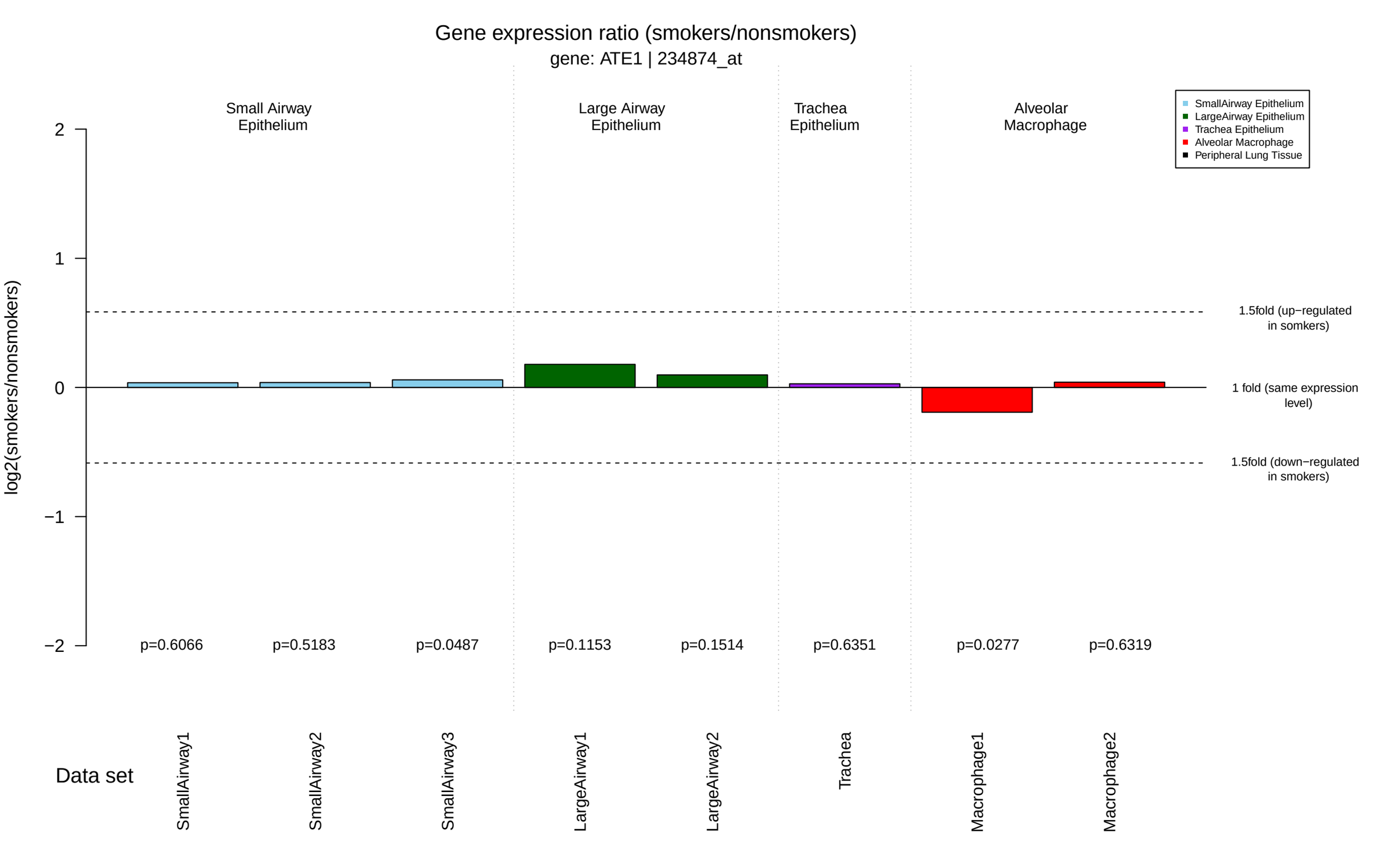
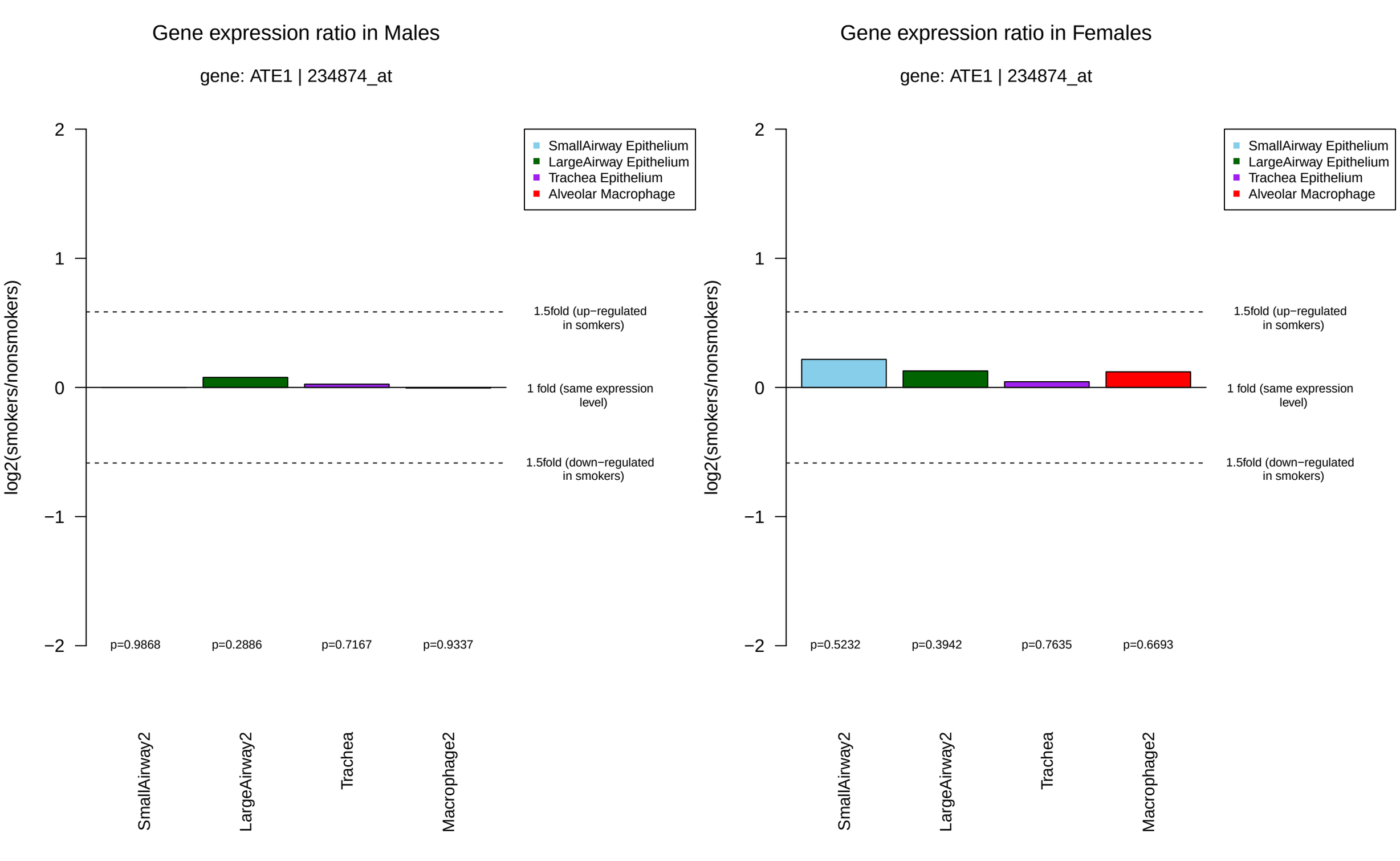
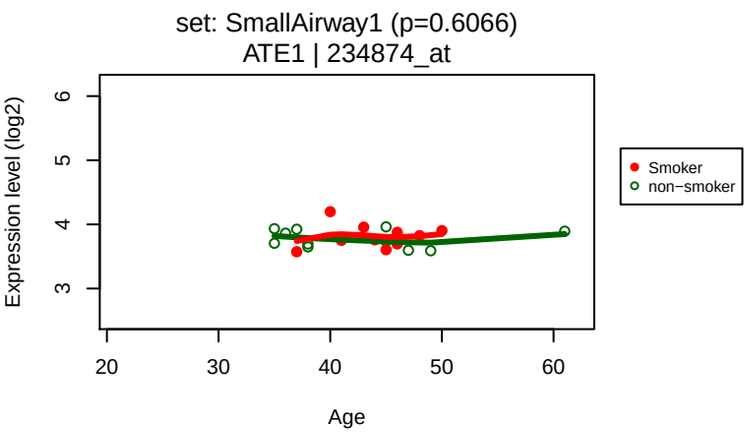
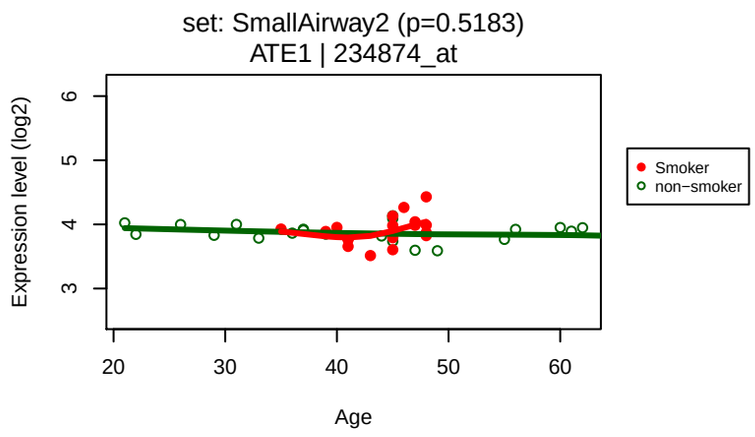
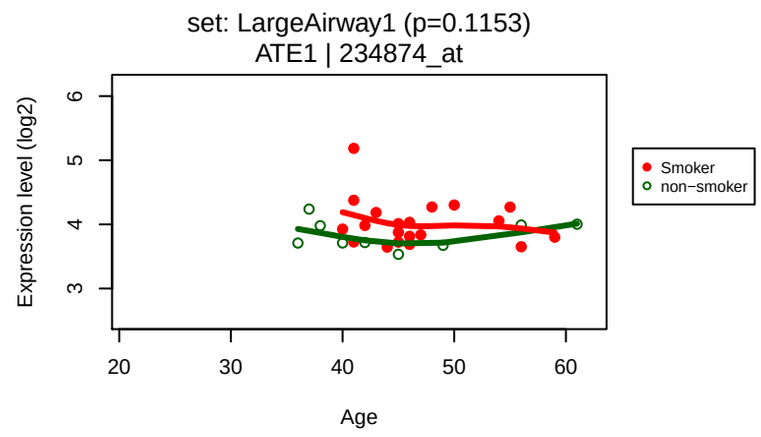

Gene expression of smokers and nonsmokers
gene: ATE1 | 234874_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6066
p=0.5183
p=0.0487
p=0.1153
p=0.1514
p=0.6351
p=0.0277
p=0.6319
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

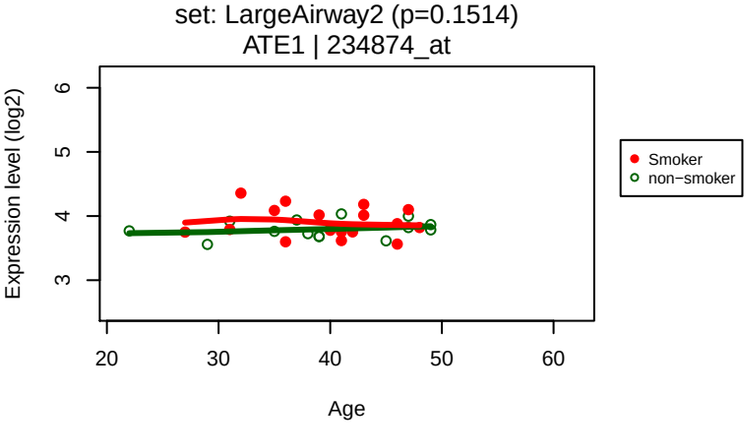

Gene expression ratio (smokers/nonsmokers)
gene: ATE1 | 234874_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6066
p=0.5183
p=0.0487
p=0.1153
p=0.1514
p=0.6351
p=0.0277
p=0.6319
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

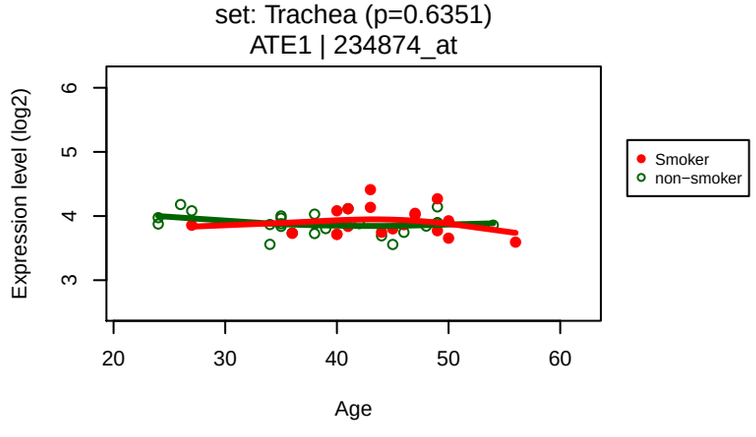

Gene expression ratio in Males
gene: ATE1 | 234874_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9868
p=0.2886
p=0.7167
p=0.9337
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATE1 | 234874_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5232
p=0.3942
p=0.7635
p=0.6693
SmallAirway2
LargeAirway2
Trachea
Macrophage2

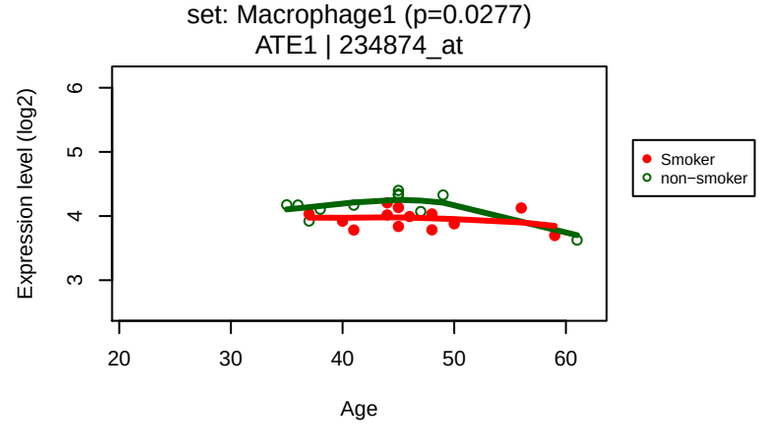

set: SmallAirway1 (p=0.6066)
ATE1 | 234874_at
Expression level (log2)
Age
Smoker
non−smoker

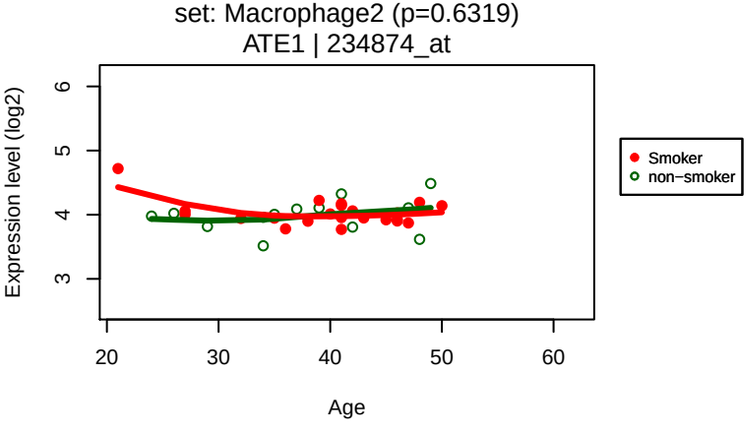

set: SmallAirway2 (p=0.5183)
ATE1 | 234874_at
Expression level (log2)
Age
Smoker
non−smoker

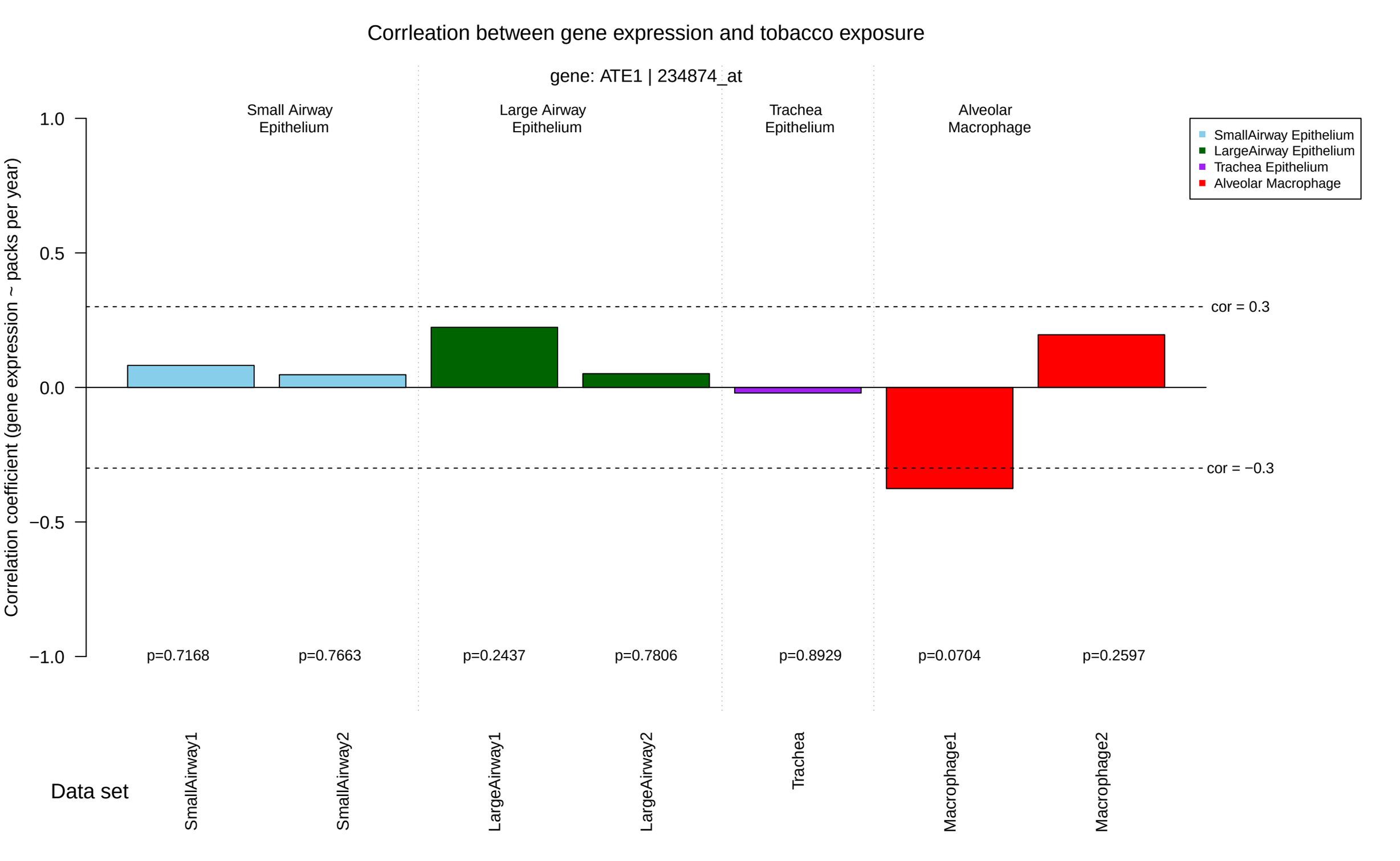

set: LargeAirway1 (p=0.1153)
ATE1 | 234874_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1514)
ATE1 | 234874_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6351)
ATE1 | 234874_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0277)
ATE1 | 234874_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6319)
ATE1 | 234874_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATE1 | 234874_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7168
p=0.7663
p=0.2437
p=0.7806
p=0.8929
p=0.0704
p=0.2597
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2